

Cell-mean fluctuations in an arithmetic convolution field: exact Rademacher-null variances and a permutation control

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Abstract

Let

$$C(N) = \sum_{n < N} \Lambda(n) \mu(N - n), \quad Z(N) = C(N) / \sqrt{V(N)},$$

where $V(N)$ is the conditional variance obtained by replacing the nonzero values of μ by independent Rademacher signs. We partition a band of even N according to how many of 3, 5, 7, 11, 13 divide N . For every cell contrast, its Rademacher-null variance is an exactly computable, doubly centred quadratic form. Its standard deviation does not follow the usual inverse-square-root cell-count scale: over eight octaves, at the three shallowest cells, it shrinks by a factor about 1.21, whereas count scaling predicts 11.3. In contrast, averaging the sign-null variance over uniformly random same-size cells gives the finite-population formula exactly. In one band, the arithmetic-cell floors exceed this permutation-averaged scale by factors from 3.8 to 105. A finite residue-class quantity derived from the pair singular series captures the floor's profile across cells. We prove that this quantity converges to a scale-free constant and compute its main term in exact rational arithmetic. Computations for a second partition and a sweep over nineteen cell shapes provide further consistency checks. No limiting law for Z , decay rate for the observed contrasts, or Goldbach consequence is asserted.

Keywords: arithmetic convolution; Möbius function; Rademacher model; permutation inference; singular series; experimental number theory.

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1 Introduction

For even N , define

$$C(N) = \sum_{n < N} \Lambda(n) \mu(N - n), \quad V(N) = \sum_{v < N} \mu^2(v) \Lambda(N - v)^2, \quad Z(N) = \frac{C(N)}{\sqrt{V(N)}}. \quad (1)$$

The field is related to a reduction used by Huang and Li in their work on binary Goldbach [1, Thm. 1, Cor. 1]; that connection supplies motivation only. Nothing below bounds $C(N)$ or yields a Goldbach consequence.

Fix a band a of even integers and partition it into cells indexed by the *depth* d , the number of primes in

$$\{3, 5, 7, 11, 13\}$$

that divide N . This choice is exploratory; computations for eighteen other prime sets are reported in Measurement 12. If m_c is the mean of Z over a cell c and $\bar{m}$ is its mean over the band, our first question is the appropriate reference scale for $m_c - \bar{m}$.

The common scale $\text{sd}(Z)/\sqrt{n_c}$, where n_c is the cell size, treats the observations as self-averaging. That assumption is inappropriate here because every $Z(N)$ reads the same underlying sequence $\mu(v)$; this is the usual design-effect issue for correlated observations [6]. Under the conditional null that preserves the support of μ^2 and replaces its nonzero signs independently, the complete covariance matrix is exactly computable. Consequently the variance of every cell contrast is also exactly computable.

The paper has three main conclusions. First, the cell floor is a three-term, doubly centred quadratic form and decreases much more slowly than cell-count scaling predicts. Second, the permutation average of the variance for a uniformly random same-size cell is exactly the usual finite-population expression. Thus the large floor is a property of the correspondence between the divisibility cells and the arithmetic kernel, not of cell size alone. Third, a residue-class main term for the pair singular series describes the profile of this floor and is asymptotically scale-free.

All standardised contrasts below use the conditional Rademacher null. This supplies an exact variance, not a Gaussian approximation or a calibrated tail law. The null preserves the observed squarefree support of μ but discards its multiplicative dependence. Thus the reported z_c values quantify departure from this specific sign model and are not calibrated probabilities for the Möbius function itself. The numerical associations involving the observed field are descriptive, and the selected prime family is not treated as a preregistered choice.

2 The exact Rademacher-null floor

We first record the elementary identity behind the calculation.

Lemma 1 (Contrast variance). *Let $X = (X_N)_{N \in a}$ be a square-integrable random vector indexed by a finite set a of size n , with covariance kernel $\Sigma_{NN'}$. For $c \subseteq a$ of size $n_c \geq 1$, put*

$$m_c = \frac{1}{n_c} \sum_{N \in c} X_N, \quad \bar{m} = \frac{1}{n} \sum_{N \in a} X_N, \quad Q_{cd} = \sum_{N \in c} \sum_{N' \in d} \Sigma_{NN'}.$$

Then

$$\text{Var}(m_c - \bar{m}) = \frac{Q_{cc}}{n_c^2} - \frac{2Q_{ca}}{n_c n} + \frac{Q_{aa}}{n^2}. \quad (2)$$

Proof. Write $m_c - \bar{m} = w^\top X$, where $w_N = n_c^{-1} \mathbf{1}_{N \in c} - n^{-1}$. Hence

$$\begin{aligned} \text{Var}(m_c - \bar{m}) &= w^\top \Sigma w \\ &= \sum_{N, N' \in a} \left(\frac{\mathbf{1}_{N \in c}}{n_c} - \frac{1}{n} \right) \Sigma_{NN'} \left(\frac{\mathbf{1}_{N' \in c}}{n_c} - \frac{1}{n} \right) \\ &= \frac{Q_{cc}}{n_c^2} - \frac{Q_{ca} + Q_{ac}}{n_c n} + \frac{Q_{aa}}{n^2}. \end{aligned}$$

Since every covariance matrix is symmetric, $Q_{ac} = Q_{ca}$, and the last display is exactly (2). $\square$

Under the sign null, write $\mu(v) = \mu^2(v)\varepsilon(v)$, where the $\varepsilon(v)$ on the squarefree support are independent Rademacher variables.

Lemma 2 (Exact cell moments). *Let a be a band of n even integers and $c \subseteq a$ a cell of size n_c . Set*

$$u_c(v) = \sum_{N \in c} \frac{\Lambda(N-v)}{\sqrt{V(N)}}, \quad Q_{cd} = \sum_{1 \leq v < N_{\max}} \mu^2(v) u_c(v) u_d(v),$$

where $\Lambda(k) = 0$ for $k \leq 0$. Under the conditional Rademacher null,

$$\text{Var}_\varepsilon(m_c - \bar{m}) = \frac{Q_{cc}}{n_c^2} - \frac{2Q_{ca}}{n_c n} + \frac{Q_{aa}}{n^2}. \quad (3)$$

The identity is exact; its reported numerical evaluations use floating-point arithmetic.

Proof. The null numerator is

$$C_\varepsilon(N) = \sum_v \mu^2(v) \varepsilon(v) \Lambda(N - v).$$

Since $E[\varepsilon(v)\varepsilon(v')] = \delta_{vv'}$ and $\mu^4 = \mu^2$,

$$\text{Cov}_\varepsilon(Z(N), Z(N')) = \frac{1}{\sqrt{V(N)V(N')}} \sum_v \mu^2(v) \Lambda(N - v) \Lambda(N' - v).$$

Therefore, for any two cells c, d ,

$$\begin{aligned} \text{Cov}_\varepsilon(n_c m_c, n_d m_d) &= \sum_{N \in c} \sum_{N' \in d} \text{Cov}_\varepsilon(Z(N), Z(N')) \\ &= \sum_v \mu^2(v) \left(\sum_{N \in c} \frac{\Lambda(N - v)}{\sqrt{V(N)}} \right) \left(\sum_{N' \in d} \frac{\Lambda(N' - v)}{\sqrt{V(N')}} \right) \\ &= \sum_v \mu^2(v) u_c(v) u_d(v) = Q_{cd}. \end{aligned}$$

Substitution of these three covariance sums into Lemma 1 gives (3). $\square$

Define the null floor and standardised contrast by

$$\text{se}_c := \sqrt{\text{Var}_\varepsilon(m_c - \bar{m})}, \quad z_c := \frac{m_c - \bar{m}}{\text{se}_c}.$$

All three terms of (3) are essential. The approximation $u_c(v) \approx n_c/\sqrt{V}$ gives the same leading contribution to each term, which cancels as $S - 2S + S$. The surviving quantity measures the departure of the cell kernel from proportionality with the band kernel.

Measurement 3 (Closed form versus simulation). In the band $(10^5, 2 \cdot 10^5]$, a simulation with 2000 independent sign draws gave

depth	0	1	2	3	4	5
n_c	19181	21097	8183	1421	115	3
closed form	0.140080	0.052724	0.167579	0.273630	0.378073	0.639330
MC/closed	0.9920	0.9993	0.9918	1.0044	1.0157	1.0320

The simulation is only a numerical check on the closed form. Its largest relative discrepancy is 3.2%, at the cell containing three points.

3 Slow decay and the failure of count scaling

On average over the v carrying weight, about $n_c/\log N$ terms in $u_c(v)$ are nonzero, each of typical size $\log N/\sqrt{V}$. This gives the heuristic

$$\frac{Q_{cc}}{n_c^2} \approx \frac{\sum_v \mu^2(v)}{V(N)} \asymp \frac{1}{\log N}, \quad (4)$$

using $V(N) \asymp N \log N$; see Appendix A. The heuristic is an averaged statement and is false at a fixed v , where parity and local congruence obstructions matter. It predicts the form of the decay, not its constant or its profile across cells.

Measurement 4 (Decay across eight octaves). Fitting $\text{se}_c \propto N^{-b}$ across the eight octaves from $(6.25 \cdot 10^4, 1.25 \cdot 10^5]$ to $(8 \cdot 10^6, 1.6 \cdot 10^7]$ gives

$$b = 0.039388, 0.039671, 0.039451$$

at depths 0, 1, 2. These values are close to the logarithmic benchmark 0.0363 over the same finite range and far from the count benchmark $1/2$. Across the factor 128 in the octave midpoints, the fitted floor therefore shrinks by about 1.21, whereas $n_c^{-1/2}$ predicts $\sqrt{128} = 11.3$.

The approximation does not describe every depth uniformly. The fitted exponent at depth 5 is 0.107306, where the cell contains only 266 points in the top octave. At that octave the exact floors from depths 0 to 5 are

$$0.1182, 0.0445, 0.1414, 0.2305, 0.3169, 0.4115.$$

They exceed $\text{sd}(Z)/\sqrt{n_c}$ by factors approximately 158, 62, 124, 84, 33, 7, respectively. Thus cell count does not describe either the scale dependence or the cross-cell profile.

For reference, the observed standardised contrasts satisfy

$$\max_c |z_c| = 10.099, 10.954, 11.179, 12.968, 11.885, 11.026, 10.019, 9.064$$

over the eight octaves. At the top octave the six depths give

depth	0	1	2	3	4	5
z_c	0.04	0.69	-0.06	-1.55	-4.46	-9.06

These are standardised descriptions; no Gaussian or multiple-comparison calibration is attached to them.

4 Permutation control

Let $K(N, N')$ denote the covariance kernel in the proof of Lemma 2. Its diagonal is exactly one because

$$K(N, N) = \frac{\sum_v \mu^2(v) \Lambda(N-v)^2}{V(N)} = 1.$$

This normalisation makes the average floor of a random cell explicit. The relabelling below is conditional on the observed field and cell sizes, as in standard randomisation inference [7].

Lemma 5 (Permutation-averaged floor). *Let c^π be a uniformly random n_c -subset of a band a of size n . Then*

$$\tau_c^2 := E_\pi \text{Var}_\varepsilon(m_{c^\pi} - \bar{m}) = \frac{n - n_c}{n_c(n-1)} \left(1 - \frac{Q_{aa}}{n^2}\right). \quad (5)$$

For the observed field held fixed,

$$\text{Var}_\pi(m_{c^\pi} - \bar{m}) = \frac{n - n_c}{n_c(n-1)} s^2, \quad s^2 = \frac{1}{n} \sum_N (Z(N) - \bar{m})^2, \quad (6)$$

and $E_\varepsilon s^2 = 1 - Q_{aa}/n^2$.

Proof. Put $J_c(N) = \mathbf{1}_c(N) - n_c/n$. Then

$$\text{Var}_\varepsilon(m_c - \bar{m}) = \frac{1}{n_c^2} \sum_{N, N'} J_c(N) J_c(N') K(N, N').$$

For $p = n_c/n$ and a uniform fixed-size subset,

$$E_\pi[J_{c^\pi}(N)J_{c^\pi}(N')] = p(1-p) \left(\delta_{NN'} - \frac{1 - \delta_{NN'}}{n-1} \right).$$

Using $\sum_N K(N, N) = n$ and $\sum_{N \neq N'} K(N, N') = Q_{aa} - n$, we obtain

$$\begin{aligned} E_\pi \text{Var}_\varepsilon(m_{c^\pi} - \bar{m}) &= \frac{p(1-p)}{n_c^2} \left(n - \frac{Q_{aa} - n}{n-1} \right) \\ &= \frac{p(1-p)}{n_c^2} \frac{n^2 - Q_{aa}}{n-1} \\ &= \frac{n - n_c}{n_c(n-1)} \left(1 - \frac{Q_{aa}}{n^2} \right), \end{aligned}$$

which is (5).

For the observed field, put $x_N = Z(N) - \bar{m}$. Then

$$\sum_N x_N^2 = ns^2, \quad \sum_{N \neq N'} x_N x_{N'} = \left(\sum_N x_N \right)^2 - \sum_N x_N^2 = -ns^2.$$

Repeating the preceding calculation with $K(N, N') = x_N x_{N'}$ gives

$$\text{Var}_\pi(m_{c^\pi} - \bar{m}) = \frac{n - n_c}{n_c(n-1)} s^2,$$

which is (6). Finally,

$$\begin{aligned} E_\varepsilon s^2 &= \frac{1}{n} \sum_N E_\varepsilon Z(N)^2 - E_\varepsilon(\bar{m})^2 \\ &= 1 - \frac{Q_{aa}}{n^2}, \end{aligned}$$

because $E_\varepsilon Z(N)^2 = K(N, N) = 1$ and $E_\varepsilon(\bar{m})^2 = Q_{aa}/n^2$. $\square$

Thus the count-based scale is recovered exactly, up to the usual finite-population correction [5], after averaging the variance over random same-size cells. It is not recovered for the divisibility cells.

Measurement 6 (Arithmetic and permuted cells). In the octave $(2 \cdot 10^6, 4 \cdot 10^6]$, the true labelling gives

depth	0	1	2	3	4	5
n_c	383617	421978	163568	28507	2263	67
z_c	0.1069	1.1133	-0.2314	-2.3990	-5.9997	-11.0258

Ten uniformly sampled fixed-count relabellings gave maxima

$$0.9359, 1.6073, 1.9921, 1.5687, 2.1392, 1.1216, 1.7178, 1.1348, 3.2040, 1.3192.$$

This small sample provides descriptive rather than precise permutation inference: the smallest attainable Monte Carlo p -value is $1/11$.

The exact comparison is between the arithmetic floor and (5):

depth	0	1	2	3	4	5
se_c/τ_c	104.80	42.68	70.28	44.37	16.97	3.83

The permutation average depends on a cell only through its size, whereas the arithmetic floor depends on which integers carry the label. The contrast therefore isolates the correspondence between the cells and the arithmetic covariance kernel.

For the fixed observed numerator, (6) also gives an exact variance without sampling. At the same octave,

depth	0	1	2	3	4	5
$ m_c - \bar{m} /\sigma_c$	11.25	47.71	16.33	106.89	102.22	42.42

where σ_c^2 denotes the right-hand side of (6). Chebyshev's inequality bounds the two-sided permutation probability for the depth-5 numerator by $5.56 \cdot 10^{-4}$ and the union of the six cellwise events by $1.28 \cdot 10^{-2}$. These bounds concern the numerator, not the ratio z_c , whose denominator changes under relabelling.

5 A singular-series descriptor

For nonzero h , let $\mathfrak{S}_2(h)$ be the Hardy–Littlewood singular series for a prime pair separated by h . Define ordered off-diagonal pair averages by

$$E_{\text{same},c}[\mathfrak{S}_2] = \frac{1}{n_c(n_c - 1)} \sum_{\substack{N, N' \in c \\ N \neq N'}} \mathfrak{S}_2(N - N'),$$

$$E_{\text{all}}[\mathfrak{S}_2] = \frac{1}{n(n - 1)} \sum_{\substack{N, N' \in a \\ N \neq N'}} \mathfrak{S}_2(N - N'),$$

and

$$D_c = E_{\text{same},c}[\mathfrak{S}_2] - E_{\text{all}}[\mathfrak{S}_2]. \quad (7)$$

The exact three-term analogue of the centred covariance form is

$$B_c = E_{\text{same},c}[\mathfrak{S}_2] - 2E_{c,a}[\mathfrak{S}_2] + E_{\text{all}}[\mathfrak{S}_2],$$

where

$$E_{c,a}[\mathfrak{S}_2] = \frac{1}{n_c(n - 1)} \sum_{N \in c} \sum_{\substack{N' \in a \\ N' \neq N}} \mathfrak{S}_2(N - N').$$

The two-term D_c is used as a descriptor, not asserted to equal the variance kernel.

Measurement 7 (Cross-cell profile). At the octave $(2 \cdot 10^6, 4 \cdot 10^6]$, a sample of 400000 pairs per cell gives the estimator $\widehat{D}_c$:

depth	0	1	2	3	4	5
$\widehat{D}_c$	0.302138	0.046766	0.412504	1.052129	1.946109	3.171298
se_c	0.12400	0.04662	0.14834	0.24178	0.33253	0.43685

The correlation across the six cells is 0.9805. This association is post hoc and descriptive. Its useful feature is not the correlation alone but the scale comparison: $\text{se}_c^2 / \widehat{D}_c$ ranges from 0.04648 to 0.06018, while $\widehat{D}_c$ itself varies by a factor about 68. After matching diagonal and centring conventions term by term, the ratio has a full width 19.8% of its mean.

We next show that D_c has a scale-free residue-class main term.

Lemma 8 (Progression sum against an envelope). *Let $w : \mathbb{Z} \rightarrow \mathbb{R}$ have finite support and total variation $\text{TV}(w) = \sum_h |w(h+1) - w(h)|$. For every $q \geq 1$ and $a \in \mathbb{Z}$,*

$$\sum_{h \equiv a \pmod{q}} w(h) = \frac{1}{q} \sum_h w(h) + O(\text{TV}(w)),$$

with an absolute implied constant.

Proof. Partition $\mathbb{Z}$ into consecutive blocks of length q , each with one representative h_j in the class $a \pmod{q}$. If I_j is the j th block, then

$$\begin{aligned} \sum_{h \equiv a \pmod{q}} w(h) - \frac{1}{q} \sum_h w(h) &= \sum_j \left(w(h_j) - \frac{1}{q} \sum_{h \in I_j} w(h) \right) \\ &= \sum_j \frac{1}{q} \sum_{h \in I_j} (w(h_j) - w(h)). \end{aligned}$$

Let V_j be the variation of w within I_j . For every $h \in I_j$, $|w(h_j) - w(h)| \leq V_j$, and hence the absolute value of the last display is at most

$$\sum_j V_j \leq \sum_h |w(h+1) - w(h)| = \text{TV}(w).$$

This proves the stated estimate with implied constant 1. $\square$

Proposition 9 (Scale-free main term). *Let $Q = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, let a be the even integers in an interval of length B , and put $\ell = B/(2Q)$. Let R_c be a nonempty set of m even residue classes modulo $2Q$, and let c be their union inside the band. Define*

$$\begin{aligned} A(h) &= \prod_{\substack{2 < p|h \\ p \leq 13}} \left(1 + \frac{1}{p-2} \right), & C_2 &= \prod_{p>2} \frac{p(p-2)}{(p-1)^2}, \\ \kappa &= 2C_2 \prod_{p>13} \left(1 + \frac{1}{p(p-2)} \right), \end{aligned}$$

and

$$\mathcal{A}_c = \frac{1}{m^2} \sum_{r, r' \in R_c} A(r - r'), \quad \mathcal{A}_a = \frac{1}{Q^2} \sum_{r, r'} A(r - r'),$$

where the second sum is over all even residue classes modulo $2Q$. Then, for every fixed $\eta \in (0, 1)$,

$$D_c = \kappa(\mathcal{A}_c - \mathcal{A}_a) + O(\ell^{-1}) + O_\eta(B^\eta/\ell) + O_\eta(B^{\eta-1}). \quad (8)$$

The constants are uniform over the finitely many depth cells and over the position of the interval. Consequently

$$D_c \longrightarrow \kappa(\mathcal{A}_c - \mathcal{A}_a).$$

If this limit is nonzero, the one-step exponent

$$e_c(B) = -\frac{\log |D_c(2B)/D_c(B)|}{\log 2}$$

tends to zero.

Proof. For even nonzero h write

$$\mathfrak{S}_2(h) = 2C_2 A(h) T(h), \quad T(h) = \prod_{\substack{p|h \\ p>13}} \left(1 + \frac{1}{p-2} \right).$$

Expanding the second product gives

$$T(h) = \sum_{d|h} g(d), \quad g(d) = \prod_{p|d} \frac{1}{p-2},$$

where d is squarefree and all its prime factors exceed 13. Every such d is coprime to $2Q$.

Write

$$f_c(h) = \#(R_c \cap (R_c + h)), \quad W_c(h) = \#\{N \in c : N - h \in c\}.$$

For each residue class counted by $f_c(h)$, the admissible N fill an interval of length $B - |h|$. Hence

$$W_c(h) = f_c(h) \frac{B - |h|}{2Q} + \rho_c(h), \quad |\rho_c(h)| \leq f_c(h) \leq m. \quad (9)$$

Also $n_c = m\ell + O(m)$, so $n_c(n_c - 1) = m^2\ell^2(1 + O(\ell^{-1}))$. The first term in (9), after normalisation, is a mixture of triangular envelopes indexed by pairs of residue classes. Each such envelope has total variation $O(B^{-1})$. Since $(d, 2Q) = 1$, the Chinese remainder theorem and Lemma 8 give

$$P(d \mid h) = \frac{1}{d} + O(B^{-1})$$

uniformly for $d \leq B$ within each admissible residue-class pair. The replacement of the exact class counts by $m\ell$ contributes $O(\ell^{-1})$. Summing the per- d discrepancy against $g(d)$ contributes

$$O\left(\frac{1}{\ell} \sum_{d \leq B} g(d)\right) = O_\eta(B^\eta/\ell).$$

The divisor errors and the tail are controlled by Rankin's trick:

$$\begin{aligned} \sum_{d \leq B} g(d) &\leq B^\eta \prod_{p > 13} \left(1 + \frac{1}{(p-2)p^\eta}\right) = O_\eta(B^\eta), \\ \sum_{d > B} \frac{g(d)}{d} &\leq B^{\eta-1} \prod_{p > 13} \left(1 + \frac{1}{(p-2)p^\eta}\right) = O_\eta(B^{\eta-1}). \end{aligned}$$

It remains to control the rounding term in (9). Since $T(h) \leq \tau(|h|)$,

$$\sum_{0 < |h| < B} |\rho_c(h)| T(h) \leq m \sum_{0 < |h| < B} \tau(|h|) = O_\eta(mB^{1+\eta}).$$

After division by $n_c(n_c - 1)$ this is $O_\eta(B^{\eta-1})$, uniformly over the finitely many cells because Q is fixed and $1 \leq m \leq Q$. Thus the primes above 13 contribute, uniformly over each residue-class pair,

$$\prod_{p > 13} \left(1 + \frac{1}{p(p-2)}\right)$$

up to the errors in (8).

Finally, $A(h)$ is constant on each pair of residue classes modulo $2Q$. Averaging those pairs over the cell and over the whole band gives $\mathcal{A}_c$ and $\mathcal{A}_a$, respectively, up to $O(\ell^{-1})$. Consequently

$$E_{\text{same},c}[\mathfrak{S}_2] = \kappa \mathcal{A}_c + O(\ell^{-1}) + O_\eta(B^\eta/\ell) + O_\eta(B^{\eta-1}),$$

and the identical calculation for the full band gives

$$E_{\text{all}}[\mathfrak{S}_2] = \kappa \mathcal{A}_a + O(\ell^{-1}) + O_\eta(B^\eta/\ell) + O_\eta(B^{\eta-1}).$$

Subtracting these two identities yields (8). Since all three error terms tend to zero for fixed $\eta \in (0, 1)$, a nonzero limiting constant gives

$$\frac{D_c(2B)}{D_c(B)} \longrightarrow 1, \quad e_c(B) = -\frac{\log |D_c(2B)/D_c(B)|}{\log 2} \longrightarrow 0.$$

□

6 Finite computations

Because $A(h)$ depends only on the residue of h modulo Q , the main term in Proposition 9 is a finite rational sum. Moreover,

$$\kappa = 2 \prod_{p \in \{3,5,7,11,13\}} \frac{p(p-2)}{(p-1)^2} = \frac{11011}{8192}. \quad (10)$$

Measurement 10 (Residue main term). The exact rational values of $\mathcal{A}_c - \mathcal{A}_a$ at depths 0 through 5 are

$$\frac{635653709}{2854051200}, \quad \frac{10743253}{313945632}, \quad \frac{318004704}{1037775739}, \quad \frac{296100736}{378194817}, \quad \frac{13823072}{9546537}, \quad \frac{78976}{33033}.$$

All are nonzero. Multiplying by (10) gives

depth	0	1	2	3	4	5
main term	0.299361	0.045996	0.411876	1.052351	1.946236	3.213542
$\widehat{D}_c$	0.302138	0.046766	0.412504	1.052129	1.946109	3.171298
residual/MCSE	2.0	0.6	0.5	0.2	0.1	260.8

Here MCSE is the pair-sampling standard error, not the null floor se_c . Four depths lie within two MCSEs; depth 0 is at the displayed boundary. At depth 5, the cell is a single residue class and contains 67 points. Every shift is $2Qk$ with $0 < |k| < 67$, so the missing tail of the divisor expansion is explicitly computable. Its contribution is 0.043558, compared with the observed residual 0.042244.

For the full residue set, the same Euler-factor calculation gives $\kappa \mathcal{A}_a = 2$; thus the limiting whole-band average of $\mathfrak{S}_2$ over even shifts is 2, consistently with the classical average of the singular series [3, 4]. Writing

$$D'_c = E_{\text{same},c}[\mathfrak{S}_2] - 2$$

removes the common sampling error in the estimated band average. With the depth-5 finite-divisor correction included, the comparison becomes

depth	0	1	2	3	4	5
D'_c	0.300893	0.045521	0.411259	1.050884	1.944864	3.170053
prediction	0.299361	0.045996	0.411876	1.052351	1.946236	3.169984
residual/MCSE	-1.13	0.36	0.45	1.07	1.16	-0.43

All entries lie within 1.2 pair-sampling standard errors. This finite-band refinement was obtained post hoc; it is an explicit calculation rather than a fitted correction.

Measurement 11 (A second cell shape). For the depth partition generated by $S = (3, 5, 7)$, the modulus is $Q' = 105$ and

$$K = 2 \prod_{p \in S} \frac{p(p-2)}{(p-1)^2} = \frac{175}{128}.$$

The pair averages were evaluated exhaustively by autocorrelation, so the row labelled exact has no pair-sampling error:

depth	0	1	2	3
m	48	44	12	1
main term	0.295736	0.126485	0.825521	2.375000
corrected lower endpoint	0.295286	0.126068	0.824967	2.374142
exact	0.295708	0.126454	0.825436	2.374142

At the three multiclass depths the exact value lies between the residue main term and the fully corrected endpoint; the single-class depth is an algebraic consistency check. The exact difference from the main term decomposes into a residue-weight error, a finite-divisor deficit, and a covariance between the low-prime factor A_S and the remaining Euler factor. No fitted parameter is required. We regard this as a second-shape consistency check, not a documented temporal holdout.

Measurement 12 (Nineteen cell shapes). The same computation was applied to every pair and triple from $\{3, 5, 7, 11, 13\}$ that the band resolves, four quadruples, the full five-prime set, and that set with 17 adjoined. Among 70 cells with $n_c \geq 1000$, every residue main term differs from the exhaustive value by less than 0.01 and by less than the magnitude of the exact value; the largest absolute difference is $2.9766 \cdot 10^{-3}$. Eight smaller cells were reported but not scored. All 70 scored differences have the same sign: the main term exceeds the exact value. This is empirical consistency across cell shapes, not a proof of the sign.

Exhaustive autocorrelation also separates the deterministic D_c from its sampled estimator. Over three consecutive octaves the relative spreads of the six exact D_c values are

$$0.0001, 0.0004, 0.0002, 0.0002, 0.0022, 0.0128,$$

and the corresponding fitted exponents are

$$-0.000059, -0.000285, -0.000136, -0.000164, -0.001584, -0.009202.$$

These finite computations agree with the scale-free conclusion of Proposition 9; they do not supply a convergence rate.

7 Computational methods and reproducibility

The computations were run with Python 3.12.10 and NumPy 2.2.5. Except for the rational residue calculations identified below, numerical sums used double precision. The repository packet contains the source, captured outputs, and SHA-256 digests. From the packet directory a run has the form

`pythoncode/<script>.py.`

The principal correspondence between results and scripts is:

Result	Implementation
Lemma 2 check	<code>lab_cellmom_montecarlo.py</code> ; 2000 sign draws
Eight-octave floors	<code>lab_cell_floor.py</code> ; exact quadratic form in double precision
Permutation control	<code>lab_mask_placebo.py</code> ; 10 fixed-count relabellings, seed 20260808
$\hat{D}_c$ profile	<code>lab_cell_singular.py</code> ; 400000 sampled ordered pairs per cell
Residue main term	<code>lab_maincoef.py</code> ; exact rational arithmetic
Second cell shape	<code>lab_secondcell12.py</code> ; exhaustive autocorrelation
Nineteen-shape sweep	<code>lab_cellset_sweep.py</code> ; exhaustive computation

The distinction between sampled and exhaustive values is maintained in the notation: D_c denotes the deterministic off-diagonal pair average, whereas $\hat{D}_c$ denotes its sampled estimator. Likewise, MCSE in Measurement 10 is a pair-sampling standard error and is not the Rademacher-null floor se_c .

The Lean 4 development formalises the algebraic identities underlying the exact cell variance and permutation formula. Its toolchain and Mathlib dependency are pinned by `lean-toolchain` and `lake-manifest.json`, and the axiom report is stored in `lean/axioms.txt`. Proposition 9, including its analytic Euler-product and error estimates, is not machine-verified.

8 Conclusion

The conditional Rademacher null makes the covariance of the convolution field explicit. The variance of a cell contrast is therefore the exact three-term form (3), rather than a count-based standard error. For the divisibility cells studied here the resulting floor is large and slowly varying, while its average over uniformly random same-size cells reduces exactly to the finite-population formula.

The excess pair singular series D_c describes much of the floor's cross-cell profile. Its residue-class main term is rational and asymptotically scale-free, and finite computations show the same mechanism across additional cell shapes. The observed association between D_c and the floor remains descriptive: no equality between the two kernels is proved, the cell family was selected exploratorily, and no limiting distribution or decay law for the observed contrasts is claimed.

Disclosure statement

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Data availability statement

Code, result files, and the Lean 4 development are available at <https://github.com/soke91/arithmetic-convolution-fields> in the directory `P4-coherent-cell-floor-r1/`. The packet contains SHA-256 hashes for the computational artifacts. Numerical results used double precision except where exact rational arithmetic is stated.

A The normalising scale

Proposition 13. *Let*

$$W(N) = \sum_{w < N} \Lambda(w)^2, \quad \mathfrak{A}(N) = \prod_{q \nmid N} \left(1 - \frac{1}{q(q-1)}\right).$$

Then

$$V(N) = W(N)\mathfrak{A}(N)(1 + o(1)) \sim \mathfrak{A}(N)N \log N.$$

Proof. Since $\Lambda(w)^2$ is supported on prime powers,

$$V(N) = \sum_{p < N} (\log p)^2 \mu^2(N-p) + O(\sqrt{N} \log^2 N).$$

For a prime $q \nmid N$, the forbidden unit class $p \equiv N \pmod{q^2}$ contributes the local factor $1 - 1/\varphi(q^2) = 1 - 1/(q(q-1))$. If $q \mid N$, the condition $q^2 \mid N-p$ forces the exceptional prime $p = q$, giving local density 1. Mirsky's theorem on squarefree shifted primes [2] therefore supplies the factor $\mathfrak{A}(N)$. This product is bounded below by the positive product over all primes, uniformly in N . More explicitly, if

$$R(N) = \sum_{p < N} \mu^2(N-p),$$

then Mirsky's theorem gives

$$R(N) = \mathfrak{A}(N) \operatorname{Li}(N) + o\left(\frac{N}{\log N}\right).$$

To insert the weight, take $\varepsilon(N) = 1/\log \log N$. The contribution of the primes below $N^{1-\varepsilon}$ is bounded by

$$(\log N)^2 \pi(N^{1-\varepsilon}) = o(N \log N).$$

On the remaining range, $(\log p)^2 = (1 + o(1))(\log N)^2$, uniformly in p . Hence

$$\begin{aligned} V(N) &= (1 + o(1))(\log N)^2 R(N) + o(N \log N) \\ &= (1 + o(1))\mathfrak{A}(N)N \log N. \end{aligned}$$

Finally, partial summation from $\theta(x) \sim x$ gives $W(N) \sim N \log N$. Dividing the last display by this asymptotic yields $V(N) = W(N)\mathfrak{A}(N)(1 + o(1))$. $\square$

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